

SAHIL SHAH

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SUMMARY

Mechanical engineering graduate with CAD proficiency across SolidWorks, AutoCAD, and Revit, and hands-on experience producing and organizing engineering documentation on active industrial projects. Familiar with M3 Engineering's project environment through a prior internship. Looking to grow into a full design and engineering role in mechanical and piping systems.

EDUCATION

B.S. Mechanical Engineering | *University of North Carolina at Charlotte*

May 2026

Relevant Coursework: Fluid Mechanics, Thermodynamics I & II, Machine Design, Solid Mechanics, Dynamics

TECHNICAL SKILLS

CAD & Drafting: AutoCAD, Revit, SolidWorks, Creo Parametric, dimensioned drawings

Engineering Analysis: MATLAB, MathCAD, Creo Simulate (FEA), fluid and thermodynamic system calculations

Documentation: Engineering drawing packages, project records, design specifications, bills of materials

Fabrication & Prototyping: FDM & SLA 3D printing, hands-on build and assembly, physical testing

Software: Microsoft 365 (Word, Excel, PowerPoint), PrusaSlicer, Arduino IDE

WORK EXPERIENCE

Engineering Intern | *M3 Engineering*

May 2024 – Aug 2024

- Organized and consolidated technical documentation packages inherited from a prior engineering firm on an active industrial project, including engineering drawings, project records, and design deliverables.
- Reviewed drawing sets to establish document control structure aligned with M3's project standards and deliverable requirements.
- Collaborated with engineering staff to ensure documentation accuracy.

Teaching Assistant — Thermodynamics II | *UNC Charlotte*

Aug 2025 – May 2026

- Supported 80+ students through organized review sessions and study materials covering thermodynamic system analysis, heat transfer, and cycle performance.
- Evaluated design exercises and homework, providing written technical feedback on student work.

ENGINEERING PROJECTS

Collegiate Wind Competition 2026 | *8-Member Multidisciplinary Team*

- Designed, fabricated, and tested small-scale wind turbine blades, applying aerodynamic simulation and Blade Element Momentum (BEM) analysis in QBlade to optimize blade geometry for target power output.
- Developed a variable-density internal blade structure, sizing shell thickness and infill density at each blade section based on expected bending loads and rotational inertia requirements.
- Built a vehicle-based field test setup to approximate wind tunnel conditions instrumented with pitot tube and independent anemometer. Correlated measured power output and tip speed ratio against simulation predictions across five wind speeds.
- Coordinated across mechanical and electrical sub-teams on system integration, test planning, and iterative design updates based on test data.

Robot Design Lab | *3-DOF Pick-and-Place Gripper Design* | *7-Member Team*

- Designed and fabricated a servo-driven gripper through three design iterations, applying force and torque analysis to set design requirements at each stage.
- Produced dimensioned CAD drawings and a full bill of materials for five fabricated components, ensuring part compatibility and design feasibility prior to fabrication.
- Used Creo Simulate for structural feasibility checks before fabrication. Validated final system performance against functional requirements through physical testing.